

ABU DHABI INTL, United Arab Emirates

WMO: 412170

Lat: 24.433N

Lon: 54.651E

Elev: 27

StdP: 101.00

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	13.2	-0.8	3.5	31.1	1.9	4.3	30.2	9.7	20.9	8.8	21.6	2.2	120	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	45.1	23.0	43.9	23.3	42.7	23.4	30.6	35.3	30.0	34.9	29.5	34.6	4.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.2	26.1	33.6	28.9	25.6	33.5	28.2	24.4	33.2	103.6	35.2	100.6	34.8	97.9	34.7	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.6	9.8	47.6	1.3	0.9	8.8	48.2	8.0	48.7	7.3	49.2	6.3	49.9		
(5)			6.9	31.6	1.8	0.7	5.7	32.1	4.6	32.5	3.6	32.9	2.3	33.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	19.4	21.0	24.0	28.3	32.4	34.4	36.3	37.0	34.2	30.4	25.5	21.2	(6)
(7)		DBStd	6.51	2.12	2.95	3.34	3.21	2.48	2.09	2.36	1.75	1.70	1.83	2.09	2.12	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	19	12	5	0	0	0	0	0	0	0	0	0	3	(9)
(10)		CDD10.0	6831	292	309	434	549	694	732	816	836	727	632	463	348	(10)
(11)		CDD18.3	3809	45	81	176	299	436	482	557	578	477	373	213	92	(11)
(12)		CDH23.3	53335	242	573	1550	3573	6446	7587	9198	9464	7281	4914	1997	511	(12)
(13)		CDH26.7	34499	42	208	691	1946	4178	5222	6721	6984	4902	2761	752	94	(13)
(14)	Wind	WSAvg	3.7	3.4	4.0	4.2	4.0	3.9	4.0	4.0	3.7	3.4	3.2	3.2	(14)	
(15)	Precipitation	PrecAvg	57	10	11	13	6	2	1	1	0	0	2	9	(15)	
(16)		PrecMax	166	52	92	59	55	35	22	21	30	2	6	25	64	(16)
(17)		PrecMin	3	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	46	16	19	17	13	7	4	5	6	0	1	6	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	34.1	38.1	41.8	44.3	45.4	46.7	44.0	40.1	35.1	31.0	(19)	
(20)			MCWB	17.7	17.5	18.5	20.2	20.8	21.9	23.3	23.2	22.3	21.6	19.5	18.8	(20)
(21)		2%	DB	27.9	31.8	35.2	39.8	42.8	43.7	45.1	45.1	42.1	38.2	33.3	29.0	(21)
(22)			MCWB	17.4	17.6	18.3	19.6	20.9	22.5	23.6	23.3	23.0	21.6	20.4	18.9	(22)
(23)		5%	DB	26.1	29.2	33.0	37.5	41.0	42.0	43.7	43.8	40.9	37.0	32.1	27.5	(23)
(24)			MCWB	16.9	17.1	18.2	19.5	21.3	22.9	24.0	23.8	23.3	21.9	20.9	19.3	(24)
(25)		10%	DB	24.1	27.0	30.8	35.2	39.1	40.2	41.8	42.1	39.2	35.7	30.8	26.1	(25)
(26)			MCWB	16.9	17.0	18.2	19.6	21.7	23.8	24.9	24.5	24.3	22.8	21.2	19.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	21.2	22.9	24.8	27.8	30.1	31.4	31.2	30.5	28.7	25.5	23.1	(27)
(28)			MCDB	24.8	26.6	28.7	30.8	33.4	34.9	35.9	36.0	34.5	32.6	30.0	26.4	(28)
(29)		2%	WB	19.6	20.2	21.9	23.8	26.7	29.2	30.6	30.7	29.7	27.8	24.2	21.8	(29)
(30)			MCDB	23.8	24.8	27.2	30.1	32.5	34.2	35.2	35.6	34.1	32.1	29.2	25.6	(30)
(31)		5%	WB	18.7	19.4	21.1	22.8	26.0	28.6	29.9	30.1	29.1	27.0	23.5	21.0	(31)
(32)			MCDB	23.2	24.3	26.5	29.7	32.1	33.9	35.0	35.0	33.8	31.8	28.4	24.9	(32)
(33)		10%	WB	18.0	18.6	20.4	22.1	25.1	27.9	29.3	29.5	28.6	26.3	22.7	20.2	(33)
(34)			MCDB	22.5	23.9	25.8	28.9	32.0	33.6	34.7	34.7	33.3	31.2	27.7	24.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	11.1	12.8	13.8	13.1	11.6	12.1	11.9	10.5	9.8	(35)	
(36)			MCDBR	12.3	13.9	14.7	16.1	16.1	15.2	13.4	13.8	13.6	13.2	12.2	11.6	(36)
(37)			MCWBR	5.5	5.7	5.8	6.4	6.6	6.7	6.9	7.0	7.1	6.4	5.3	5.3	(37)
(38)		5% WB	MCDBR	10.2	10.6	11.2	13.1	14.4	13.0	11.4	11.8	12.0	11.7	10.7	10.2	(38)
(39)			MCWBR	5.6	5.5	5.4	5.9	6.2	6.0	5.9	6.0	5.9	5.7	5.1	5.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.449	0.502	0.569	0.605	0.592	0.634	0.727	0.700	0.598	0.506	0.458	0.439	(40)	
(41)		taud	2.017	1.869	1.720	1.673	1.688	1.643	1.529	1.578	1.728	1.905	2.030	2.050	(41)	
(42)		Ebn at Noon	791	775	746	731	737	702	639	654	713	761	775	781	(42)	
(43)		Edh at Noon	155	193	235	251	246	256	287	273	230	183	152	145	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.22	4.98	5.86	6.68	7.40	7.52	7.04	6.82	6.36	5.57	4.53	3.97	(44)	
(45)		RadStd	0.23	0.38	0.39	0.39	0.20	0.15	0.23	0.14	0.13	0.09	0.11	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.53	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	+189	+186
(47) Regional (9 neighbors)		+0.42	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	+150	+149

Nomenclature: See separate page